

Diagnosing Conformal Prediction Failures Under Distribution Shift

A COVID-19 Case Study

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Top-1 SHAP Concentration Flags GBM CP Risk Pre-deployment signal for multiclass APS conformal prediction.

$\rho = 0.853$
16-task correlation

40%
exploratory cutoff

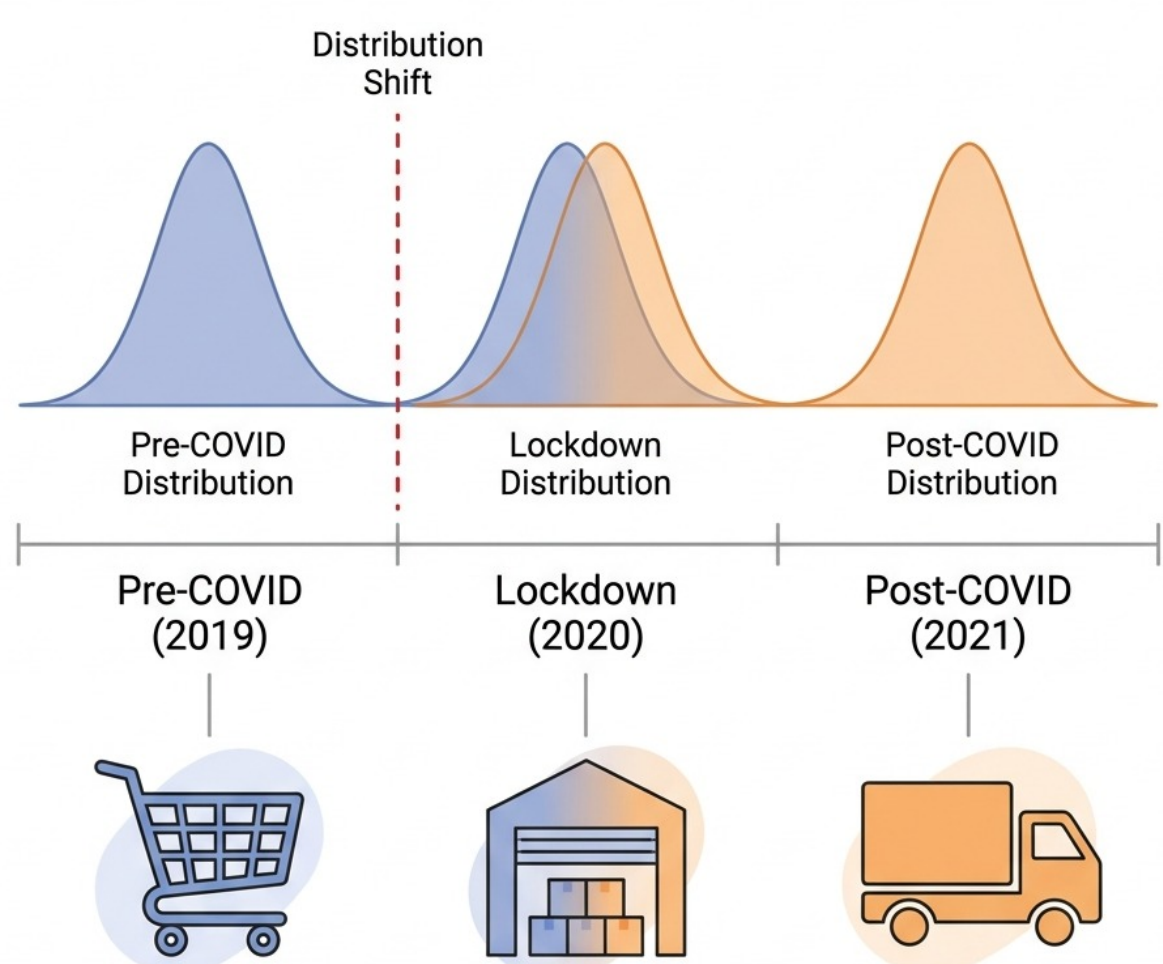
91.7%
12-check audit

Problem & Motivation

Conformal Prediction (CP):
distribution-free uncertainty.
Problem: shift can break coverage.

Research Question:

Can we **predict** when CP fails **before** deployment?



Key Insight

Models with **concentrated feature reliance** are at risk:

- Single feature drift $\Rightarrow$ large impact
- Less redundancy across features
- APS scores inflate after shift

High concentration

Risk flag

SALT Case Study & Analysis Set

RelBench Supply Chain Benchmark

- 8 classification tasks
- COVID-19 temporal split
- Feb–Jul 2020 (onset $\rightarrow$ peak)
- LightGBM multiclass APS conformal prediction

Primary correlation adds 8 external multiclass tasks, yielding 16 tasks across 9 domains.

Why Not Standard Shift Tests?

MMD/C2ST/PSI detect dataset shift.
They do not say which CP model will lose coverage.

$\rho \leq 0.19$

weak relation to SALT coverage drop

Interpretation: SHAP concentration asks whether one fragile model direction already dominates.

MAIN RESULT

Pre-deployment risk signal predicts coverage drop

LightGBM/GBM multiclass APS conformal predictors under shift

$\rho = 0.853$
Spearman Correlation

$p < 0.001$ | $n = 16$ tasks | 9 domains

Pre-deployment SHAP Concentration vs. Coverage Drop

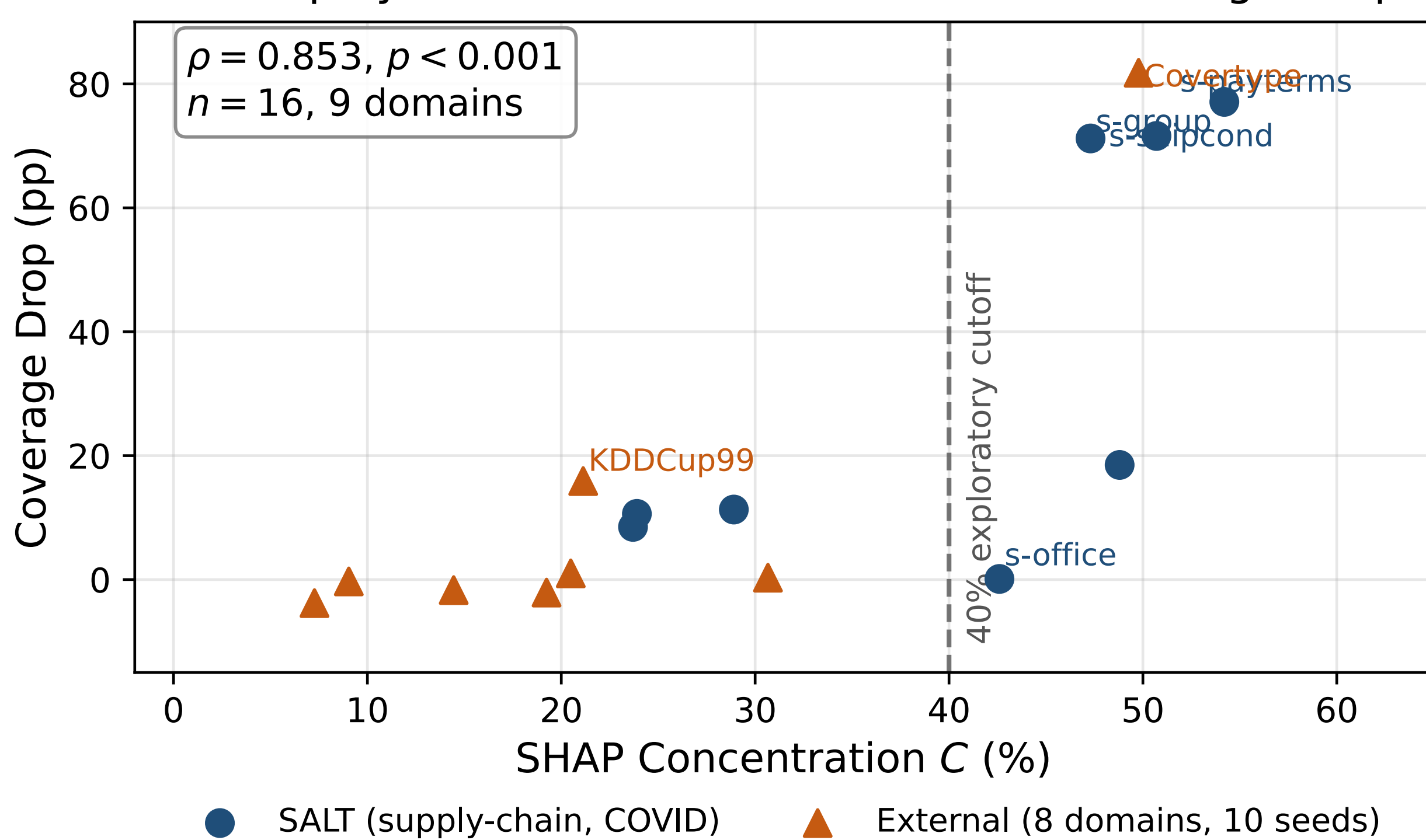


Fig. 1: Each point is a task. C is measured before deployment; coverage drop is validation-to-post-shift loss in percentage points. The dashed line is the exploratory SALT-derived cutoff.

Takeaway: higher concentration tracks larger CP coverage loss.
Across 16 tasks, high- C models occupy the high-drop region.

Mechanism & Theory

Top-1 concentration: share of mean absolute SHAP importance

$$C = \frac{\max_j \mathbb{E}|\phi_j|}{\sum_{j=1}^p \mathbb{E}|\phi_j|}$$

Idealized theorem: concentration amplifies APS score inflation

$$\mathbb{E}[s_{\text{test}}] \geq C(1 - (K-1)\epsilon) + (1-C)(1 - (K-1)\bar{h})$$

If the dominant shifted feature gives weaker true-class support ($\epsilon < \bar{h}$), larger C puts more weight on that damaged signal. APS test scores rise while the CP cutoff stays fixed, so coverage drops.

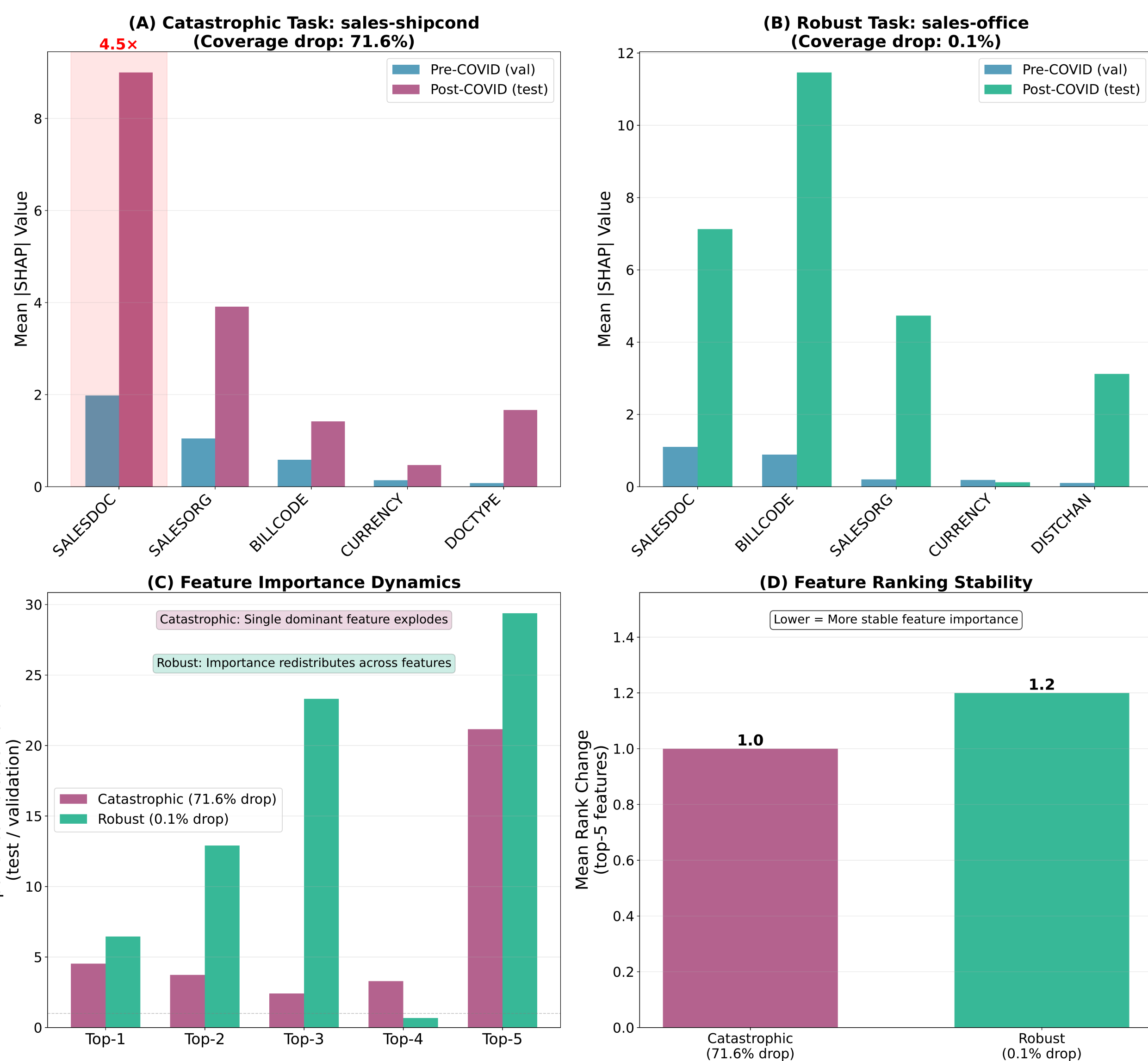


Fig. 2: Catastrophic tasks concentrate SHAP mass on one shifted feature; robust/protected tasks keep reliance distributed or stable.

A. **Failure:** SHAP mass jumps to one feature.
C. **Ratio:** failing task has a top-feature spike.

B. **Robust:** importance stays distributed.
D. **Stability:** robust tasks keep feature order.

Mechanism: high- C GBMs can create a single point of failure.

The claim is directional: stable/protective top features can be exceptions.

Specificity Check: Low- C Holds Out

Low- C checks avoid catastrophic loss.

Negative control: below-cutoff tasks should remain low risk.

Dataset	Top-1 C	Drop	Flag
CivilComments	26.2%	-0.7pp	Low risk
Amazon	1.9%	+0.4pp	Low risk
Covertypes*	12.5%	+8.0pp	Low risk
Adult	27.5%	+1.4pp	Low risk

Drop = coverage loss in pp; * different split/protocol from Fig. 1 Covertypes.

91.7%

11/12 rule labels matched
8 SALT + 4 specificity checks

Conclusions & Future Work

Takeaways:

- Top-1 SHAP concentration predicts APS CP coverage loss in GBMs: $\rho = 0.853$ ($p < 0.001$)
- The 40% cutoff is exploratory: **91.7%** label agreement across 12 checks (11/12)

Future Directions:

- **Prospective validation:** test the cutoff on new domains
- **Protective factors:** detect stable top features that prevent failure
- **Neural diagnostics:** gradient-based explanations for MLPs
- **Mitigation:** reduce concentration via features, dropout, ensembles

Scope & Model Sensitivity

Model	ρ	Why
GBM	0.55–0.83	SALT-only model-family sensitivity; the headline $\rho = 0.853$ is the 16-task LightGBM endpoint.
RF	0.30	Bagging dilutes single-feature dependence.
MLP	0.43	Failures may be distributed/global; needs neural diagnostics.

Scope: GBM multiclass APS CP. SHAP concentration captures concentrated-dependence failures, not every model failure mode.

Open Questions

- Can we design models that are **intrinsically low-concentration**?
- Does the **40% threshold** hold prospectively across new domains?
- When does high concentration stay safe because the top feature is stable?
- Can **streaming CP** monitor concentration and trigger mitigation before failure?

Paper, Code & Contact

Repository

github.com/ChorokLeeDev/
conformal-covid-uai2026

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Scan for paper & code